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DC Motor RPM Display On Smartphone Using Arduino

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This project presents an Arduino based speed monitoring system where the RPM (revolutions per minute) of a DC motor is continuously displayed on a smartphone using Bluetooth communication.

The Arduino programming employs an interrupt service routine to determine the motor speed. The authors' prototype of the project is shown in Fig. 1.

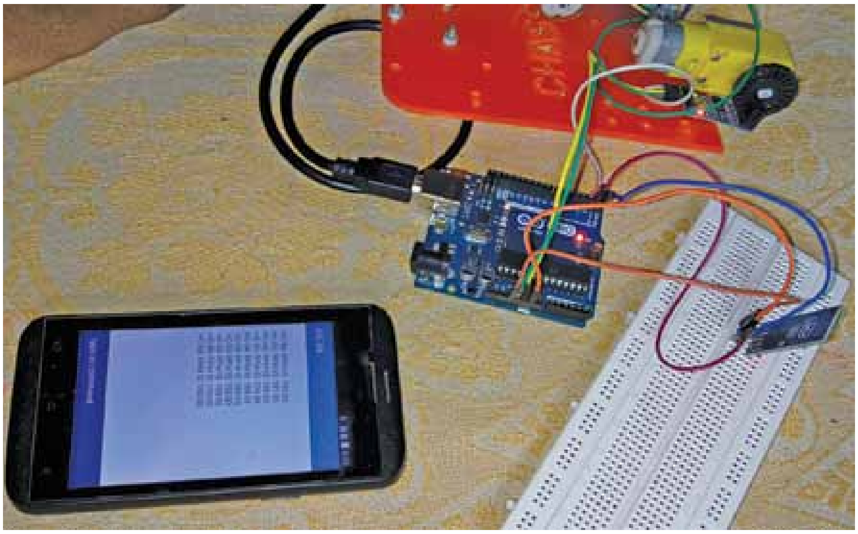

Fig. 1: Authors' prototype of the speed monitoring system

Circuit and working

The main components used in this project are:

Arduino Uno. Arduino Uno is an AVR ATmega328P microcontroller based development board with six analogue input pins and fourteen digital I/O pins. The microcontroller has 32kB ISP flash memory, 2kB RAM, and 1kB EEPROM.

Speed sensor. In this project, an opto-coupler based speed sensor module is used. This type of sensor generates square wave of a certain frequency which depends on the speed of motor. Opto-coupler module FC-03 was used during testing.

Coded disc encoder. It has some holes which help the opto-coupler type of speed sensor to generate the square wave as mentioned above. A coded disc encoder with twenty holes is used in this project.

Bluetooth module. The popular HC-06 Bluetooth module used for this project sends the data from Arduino Uno to the Android based smartphone. You can also use HC-05 Bluetooth module in place of HC-06.

Arduino Bluetooth controller. This software is based on the Android platform and helps in communication of the smartphone with the HC-06 Bluetooth module. Arduino Bluetooth Controller by Giumig Apps was used during testing.

The internal circuit diagram of the opto-coupler type speed sensor module (FC-03) is shown in Fig. 2. The circuit diagram for the monitoring of the motor speed on smartphone using Arduino Uno is shown in Fig. 3.

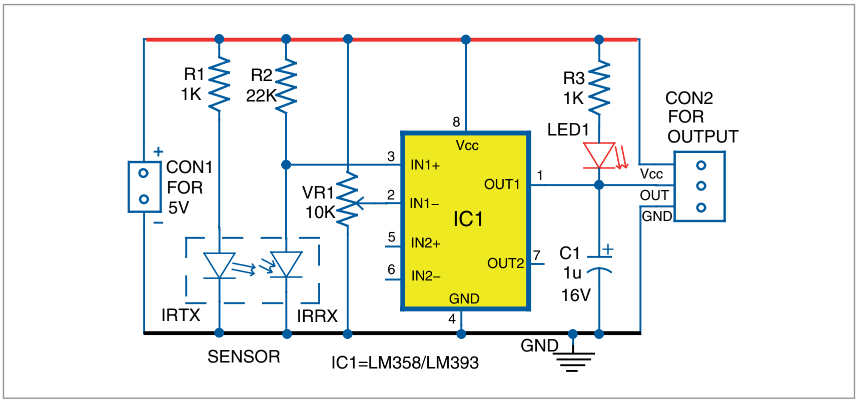

Fig. 2: Internal circuit diagram of Opto-coupler type speed sensor module (FC-03)

As shown in Fig. 3, the output of the speed sensing module is connected to digital pin 2 (D2) of the Arduino. Using an interrupt based approach, we can detect the pulse produced by the sensing module (in every ten seconds), which depends on the speed of the motor. Since there are twenty holes in the coded disc, the speed (in RPM) = ((pulse × 6)/20).

After determining the motor speed, the Arduino sends this data to the smartphone (to display) via the Bluetooth module. Pin TX of the Arduino is used for this purpose.

Software

The source code is written in Arduino programming language. The ATmega328/ATmega328P is programmed using the Arduino IDE